

STRATEGIC HEALTH INSIGHTS

U.S. County-Level Health Outcomes Analysis

Driving Telehealth Entry Strategy & Community Consulting Opportunities

Agenda

01

Health Outcome Landscape

Key drivers of life expectancy, physical & mental health across 3,143 counties

02

The Access Gap

How insurance coverage and provider availability shape outcomes, even controlling for income

03

Telehealth Opportunity Map

Identifying 477 priority counties for physical & mental telehealth services

04

Consulting Focus Framework

Four county archetypes, 79% of high-burden counties concentrated in the South

05

Methodology & Recommendations

Data-driven entry priorities and next steps

01 | Health Outcome Landscape

3,143

U.S. Counties Analyzed

75.8 yrs

Avg. Life Expectancy

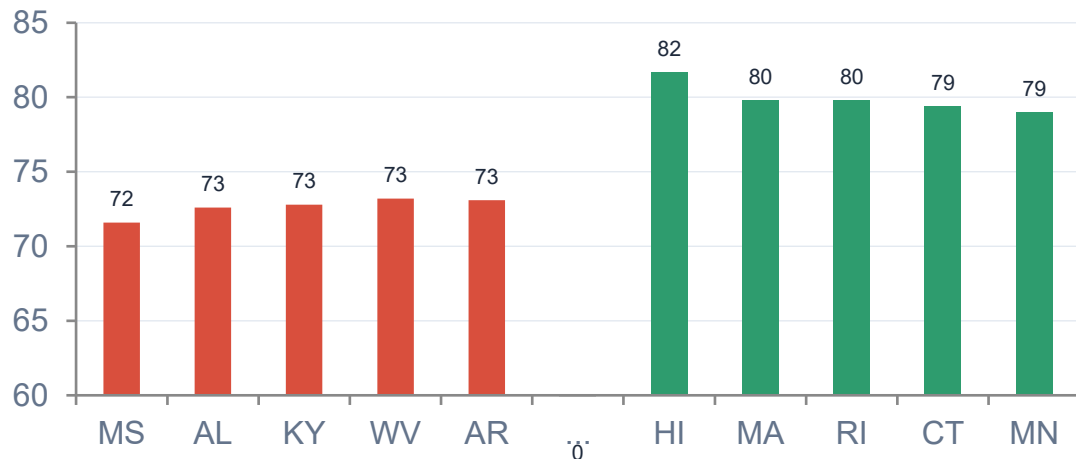
3.9 days

Avg. Poor Physical
Health Days/Month

5.2 days

Avg. Poor Mental
Health Days/Month

Life Expectancy by State — Worst vs. Best



Key Finding

10-year gap in life expectancy separates best from worst states.

This gap is larger than the mortality impact of most chronic diseases, and it's driven by measurable, addressable factors.

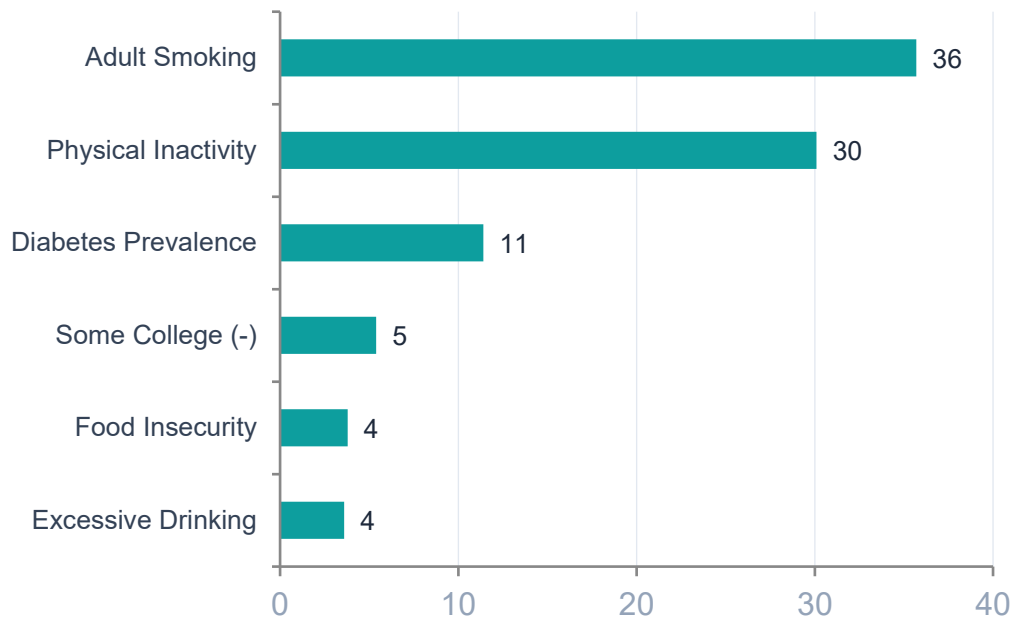
Implication: The gap is preventable, not inevitable.

Data: County Health Rankings 2024 (n=3,143 counties). Outcomes: Life Expectancy, Poor Physical Health Days, Poor Mental Health Days.

01 | What Drives Health Outcomes?

Top predictors of Poor Physical Health Days across 3,143 counties:

Feature Importance Score (%)



Method: Random Forest Regressor (sklearn), 5-fold CV. Variables ranked by Gini importance. Physical health model $R^2=0.81$ (test $R^2=0.89$); mental health $R^2=0.57$.

Random Forest Model | $R^2 = 0.81$

Mental Health Days, Key Drivers ($R^2=0.57$)

44%

Adult Smoking

Behavioral & socioeconomic overlap

14%

Food Insecurity

Chronic stress from access gaps

9%

Insufficient Sleep

Bidirectional link with mental illness

Implication for GBH

Behavioral & socioeconomic variables explain a majority of health outcome variation

Healthcare access plays a role, but is secondary to root causes

Telehealth alone is not enough, must combine with behavioral & economic interventions

02 | The Access Gap — Does It Matter Beyond Income?

Life Expectancy

High PCP Access: 76.6 yrs

Low PCP Access: 74.8 yrs

+1.8 yrs

Poor Physical Health Days/Mo

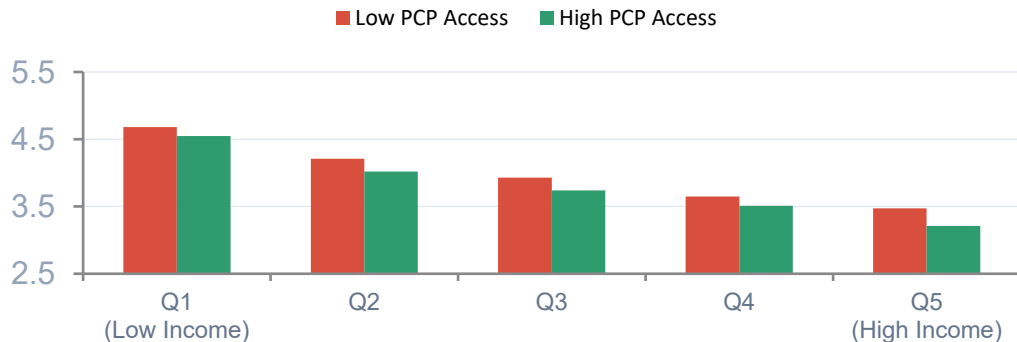
High PCP Access: 3.70 days

Low PCP Access: 4.11 days

-0.41 days

Poor Physical Health Days by Income Level & PCP Access

Access gap persists at every income level



Access Gaps at a Glance

1.8 years of life expectancy
lost in low-access counties

Even after controlling for socioeconomic variables, PCP access remains a significant independent predictor of health outcomes.

Access is necessary, but not sufficient. Its impact is smaller than behavioral and socioeconomic drivers.

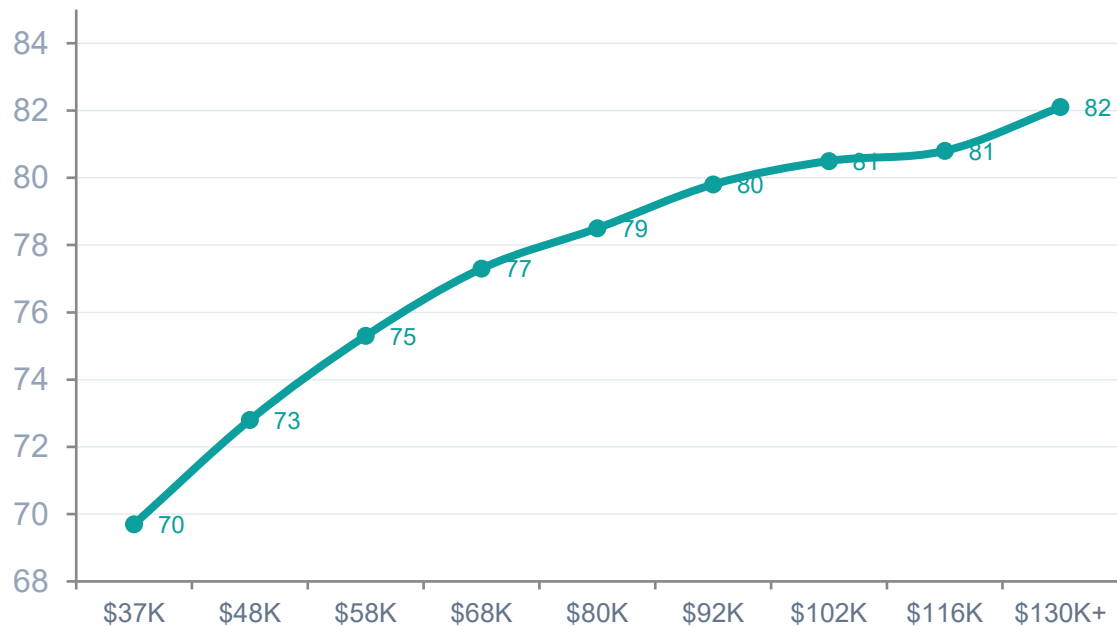
Access effect is not explained away by income, the grouped chart shows the PCP gap persists across all five income quintiles.

02 | Income & Life Expectancy — The Data Story

Each data point represents the average life expectancy for counties at a given income level (n=3,070 counties):

Median Household Income vs. Life Expectancy

(County averages, binned by income group)



Strong Relationship

$r = 0.71$

Income explains 50% of life expectancy variance across counties.

But income isn't destiny.

Counties with similar incomes show up to 8 years of variation in life expectancy, driven by behavioral and access factors GBH can influence.

Implication for GBH

Economic consulting + telehealth = dual levers on the two biggest drivers of life expectancy.

Correlation (income vs. LE): $r=0.71$. Income explains 50% of variance. Remaining 50% driven by behavioral factors and access — the exact space GBH operates in.

03 | Telehealth Opportunity - 477 Priority Counties

265 Counties

Physical Telehealth Priority
Rural + Low PCP + High Physical Burden

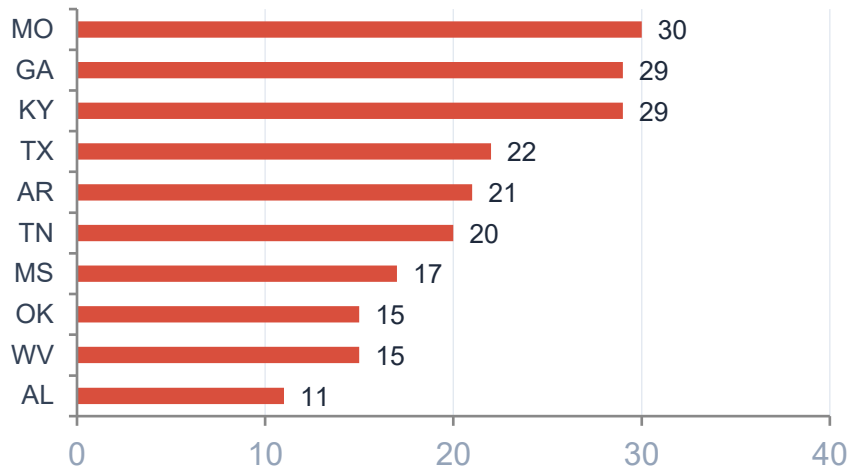
212 Counties

Mental Health Telehealth Priority
Rural + Low MHP + High MH Burden

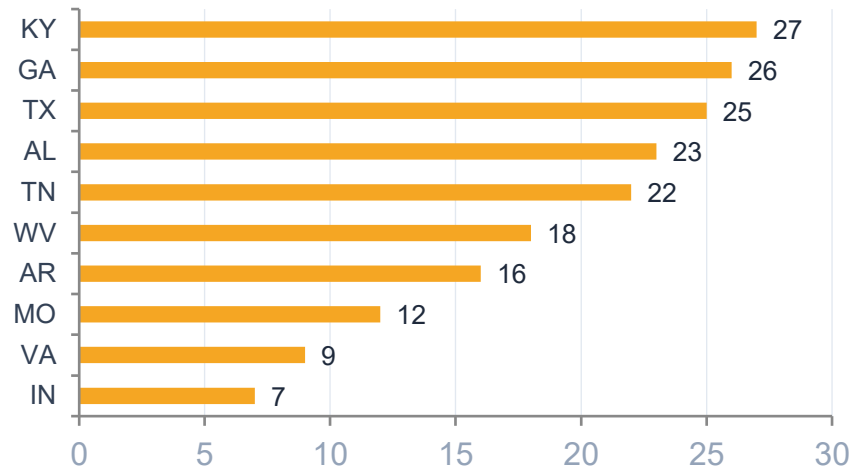
79%

High-Burden Counties
Located in Southern States

Physical Telehealth: Top Target States



Mental Health Telehealth: Top Target States



These 477 counties represent the highest ROI entry point for GBH's telehealth expansion, where access gaps AND health burden overlap, not just rural geography.

04 | Four County Archetypes - Consulting Targeting

K-means cluster analysis of 3,000+ counties on 11 health variables reveals four distinct profiles:

A

High Performers

526 counties • LE: 79.1 yrs • CO, VA, CA, MN, WA

Smoking 14.4%, poverty 11.6%, income \$82,975

GBH: Preventive wellness & benchmarking

B

Moderate: Rural Midwest

719 counties • LE: 77.5 yrs • IA, MN, WI, NE, KS

Smoking 17.6%, poverty 13.9%, income \$69,405

GBH: Telehealth access + chronic disease prevention

C

High Burden: Deep South

559 counties • LE: 72.1 yrs • TX, MS, GA, KY, AR

Smoking 23.9%, poverty 30.5%, income \$46,763

GBH: Economic dev + chronic disease management

D

Moderate Burden: Diverse

955 counties • LE: 74.7 yrs • TX, MI, NC, IL, IN

Smoking 19.9%, poverty 20.8%, income \$58,070

GBH: Behavioral health + workforce programs

Cluster C = highest ROI consulting opportunity: 559 counties, LE gap of 7 years vs. Cluster A, concentrated in accessible Southern geography.

04 | Consulting Framework - Four Root Cause Pillars

Evidence-backed focus areas for GBH's consulting arm, each pillar tied directly to model drivers:

1 Behavioral Health

ROOT CAUSES

Smoking (35.7% model weight)
Physical inactivity (30.1%)
Insufficient sleep (8.7%)

GBH SOLUTION

Smoking cessation, community wellness, corporate health partnerships

35.7% of physical health variance

2 Economic & Food Security

ROOT CAUSES

Child poverty ($r=-0.75$ with LE)
Food insecurity (14.2% mental weight)
Income inequality

GBH SOLUTION

SNAP access partnerships, workforce development, economic mobility programs

$r = -0.73$ with Life Expectancy

3 Healthcare Access

ROOT CAUSES

Uninsured rate (#1 access driver)
Provider deserts
Preventable hospitalizations

GBH SOLUTION

Insurance enrollment drives, telehealth deployment, care navigation

1.8-year life expectancy gap

4 Education & Human Capital

ROOT CAUSES

College attainment ($r=-0.73$)
HS completion ($r=-0.69$)
Disconnected youth

GBH SOLUTION

Health literacy programs, community college partnerships, youth interventions

$r = -0.73$ with health days

Appendix | Methodology

Data Source: County Health Rankings & Roadmaps 2024, University of Wisconsin Population Health Institute | n = 3,143 U.S. counties | 88 variables

Outcomes: Life Expectancy (v147), Poor Physical Health Days (v036), Poor Mental Health Days (v042)

Correlation Analysis

Pearson r computed for 20+ predictors against each outcome variable. Used to identify the strongest individual-variable relationships and screen for multicollinearity.

Result: Smoking, inactivity, poverty among top correlates ($|r| > 0.70$)

Random Forest Regression

Ensemble model (sklearn, 100 trees) trained on 20 predictor variables. 5-fold cross-validation used to estimate out-of-sample R^2 . Feature importances from mean Gini decrease.

Result: Physical health $R^2=0.81$ (CV) / 0.89 (test) | Mental health $R^2=0.57$

K-Means Clustering

K-means ($k=4$, 10 random starts) applied to 11 standardized health and socioeconomic variables. Cluster count selected by visual elbow inspection and interpretability.

Result: 4 archetypes explaining county-level health variation

All analysis in Python (pandas, scikit-learn). Note: Results reflect associations from observational county-level data, not causal relationships. Findings are directionally robust across methods.

05 | Strategic Recommendations

TELEHEALTH - PHYSICAL

Launch in the Southern access desert corridor

- Target MO, GA, KY, TX, AR, TN - 265 priority counties
- Focus on chronic disease mgmt (diabetes, obesity, smoking)
- Partner with county health depts for trust & reach

TELEHEALTH - MENTAL HEALTH

Deploy MH platform across Appalachian & Gulf states

- Priority: KY, GA, TX, AL, TN, WV - 212 counties
- Address smoking/food insecurity co-morbidities
- Use async/text services for rural broadband gaps

CONSULTING - NEAR TERM

Build practice around Clusters C & D first

- 1,500+ counties, highest burden, strongest ROI signal
- Lead with behavioral health & chronic disease programs
- Deep South (Cluster C) = 7-year LE gap vs. top counties

CONSULTING - POSITIONING

Differentiate on data-driven methodology

- Lead with the 4-pillar framework as proprietary IP
- Use county cluster profiles in sales engagements
- Target health officials, hospital networks, state agencies

Health inequity is concentrated, predictable, and addressable, 79% of high-burden counties are in the South. GBH can lead with data and win with execution.

The Opportunity Is Clear.

3,143 counties. 79% burden concentrated in the South. Four evidence-backed levers.

- **Behavioral & socioeconomic factors explain a majority of health outcome variation** → Smoking cessation & physical activity programs are the highest-leverage consulting entry point
- **Access matters beyond income** → PCP gap accounts for 1.8 life years even after controlling for socioeconomics
- **477 counties need telehealth now** → Concentrated in MO, GA, KY, TX, AR, TN, WV - Southern launch is the clear first move
- **4-pillar framework = defensible IP** → Behavioral, Economic, Access, Education, tied to model output, not just intuition

GetBetterHealth.com | MBA 551 | County Health Rankings 2024 | n=3,143 counties

The opportunity is not just to expand access, but to systematically improve health outcomes at scale.